INCH-POUND
MIL-DTL-24643/27E
22 August 2002
SUPERSEDING
MIL-C-24643/27D
22 November 1994

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, 600 VOLTS, TYPE LS2AU (INCLUDING VARIATION LS2AUS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of MIL-DTL-24643 listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation.

REQUIREMENTS:

Qualification required.

Construction (non-watertight)

First - Copper conductor, tin coated, ASTM B 286, size 22-7.
Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third - Cross-linked polyethylene insulation. Minimum average wall thickness 0.024 inch. Telephone identification code applied by method 3.
Fourth - Two conductors cabled together to form a pair, with nominal lay of 1-1/2 inches.
Fifth - Forty pairs cabled together with a lay not greater than 16 times the overall diameter of the layer. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward.
Sixth - Binder tape, applied helically with overlap.
Seventh - Braided shield of tin coated copper.

Construction variant for type LS2AU

Eighth - Binder tape, applied helically with overlap.
Ninth - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Construction variant type LS2AUS

Eighth - Spirally wrapped 0.002 inch thick polyethylene (polyester film) separator tape with 25 percent minimum overlap.
Ninth - Braided shield of tin coated copper.
Tenth - Binder tape applied helically with overlap.
Eleventh - Cross-linked polyolefin jacket. Minimum average wall thickness 0.060 inch.

TABLE I. Details.

Military part no. 1/ M24643/27	Size 2/	Number of pairs	Diameter over shielded core (inch nom.)	Overall diameter (inches)			
				TYPE LS2AU		TYPE LS2AUS	
				min	max	min	max
-01U0	40	40	1.089	1.320	1.370	1.380	1.430

1/ When double shield is required, see MIL-DTL-24643 for configuration of part number.
2/ Type LS2AU and LS2AUS.

EXAMINATION AND TESTS:

Basic electrical:	Requirements:
Conductor resistance - ohms/1000 feet at 25°C, maximum	17.71
Voltage withstand - volts, root mean square, minimum	
Conductor to conductor	2500
Shield to shield (type LS2AUS only, overall shields only)	200
Conductor to shield	1250
Insulation resistance - megohms -1000 feet, minimum	
Conductor to conductor	5000
Shield to shield (type LS2AUS only, overall shields only)	100
Jacket flaws	No failure
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Capacitance - at 1 MHz, pF/FT maximum	
Mutual capacitance	20
Group B:	
Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - at 0 ± 2°C, mandrel diameter	
12 times cable diameter	No damage
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Elongation - percent, minimum 250
Jacket (cable)
Tensile strength - lb/in^2, minimum 1300
Elongation - percent, minimum 160

Group C:
Physicals - aged
Insulation
Tensile strength - percent of unaged, minimum 80
Elongation - percent of unaged, minimum 80
Jacket (cable)
Air oven
Tensile strength - percent of unaged, minimum 60
Elongation - percent of unaged, minimum 60
Hot oil immersion
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged, minimum 50
Shrinkage ..... No failure
Heat distortion - percent of unaged, maximum 30
Permanence of printing (jacket) - cycles, minimum 125

Shield - conformance to material, construction and coverage ..... No failure
Surface transfer impedance (type LS2AUS only)
Milliohms per meter, maximum 700
EMP response time - dB, minimum 60

Group D:
Flame propagation (cable) ..... No failure

QUALIFICATION INSPECTION:

Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

Aging and compatibility (cable) (125 ± 5°C.) ..... No failure
Abrasion resistance (jacket) - scrapes, minimum 75
Acid gas equivalent - percent, maximum
Jacket 2
Fillers 2
Insulation 18
Halogen content - percent, maximum
Jacket 0.2
Fillers 0.2
Immersion (jacket)
Tensile strength - percent of unaged, minimum 50
Elongation - percent of unaged minimum 50
Smoke index, maximum
Jacket 25

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.

Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTH:
1000 feet nominal.

Custodians:
Army - MI
Navy - SH

Preparing Activity:
Navy - SH
(Project 6145-2308-025)

Review Activities:
Army - AV, CR
Navy - CG, EC
DLA – CC

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:04Z
Check the source to verify that this is the current version before use.